Types of Analytics

BY THE END:

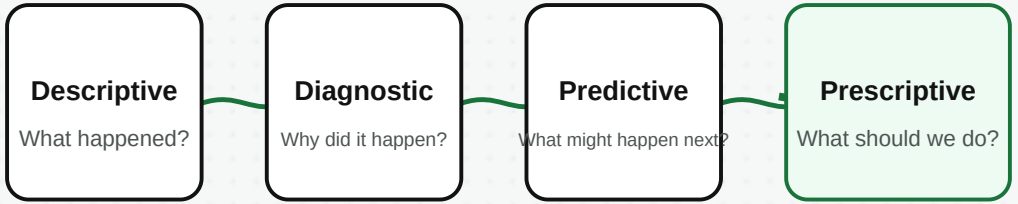
You can tell descriptive, diagnostic, predictive, and prescriptive questions apart.

Overview

A manager looks at last month's sales report and asks three questions in one meeting: "What happened? Why did it happen? What should we do next?" Each question needs a different kind of analysis.

What is analytics?

Analytics means using data to understand what happened, why it happened, what may happen next, or what action to take.



The four types of analytics

TYPE	QUESTION	EXAMPLE
Descriptive analytics	What happened?	Rice sales rose 12% last month
Diagnostic analytics	Why did it happen?	They rose because the Springfield shop stayed open longer
Predictive analytics	What might happen next?	Sales may stay high next month if the longer hours continue
Prescriptive analytics	What should we do?	Keep the longer hours in Springfield, and try them in Riverton

Simple way to remember them

Descriptive = report the past

Diagnostic = explain the past

Predictive = forecast the future

Prescriptive = recommend action

Most beginners start with descriptive analytics, such as summaries, totals, and charts. More advanced work adds prediction and recommendation.

Why this matters

The four often build on one another, and that is the useful part. You cannot explain why rice sales rose until you know that they rose. Understanding what drove last month can strengthen a forecast of next month. And a recommendation is more trustworthy when the earlier questions have supplied evidence for it.

So when an organization says it wants prediction, the honest first question is whether it can yet describe what already happened.

